MIL-C-22931/9
10 February 1968
SUPERSEDING
MIL-C-22931/1A (in part)
27 March 1964

MILITARY SPECIFICATION SHEET

CABLES, RADIO FREQUENCY

SEMIRIGID, COAXIAL, SEMI-AIR-DIELECTRIC,

.500 AND .530 INCH OUTSIDE DIAMETER, 50 OHMS

The complete requirements for procuring cables described herein shall consist of this document and the latest issue of Specification MIL-C-22931.

REQUIREMENTS:

Operating temperature range: -55 to +80° C.

Inner conductor: See table.

Dielectric:
Cable core and insulating layer: See table.

Outer conductor: Aluminum tubing, smooth, seamless.
Diameters - See table.

Jacket: See table.

Voltage withstand: 1,800 volts rms (minimum) at 60 Hz, or
3,200 volts dc

Insulation defects, spark: See table.

Attenuation: See table.

Velocity: See table.

Capacitance: 24.0 pf/ft.

Impedance: 50 ±2 ohms.

Voltage standing wave ratio:

Frequency range	Initial	After bending and temperature cycling
(MHz)	(max)	(max)
500 - 2,000	1.15	1.18
2,000 - 5,000	1.20	1.25
5,000 - 10,000	1.20	1.25

(For -001 and -002, half wavelength is at 9,300)

Mandrel size for temperature cycling, bending, and cold bend tests: 15 inches (381.0 mm).

Minimum bending radius: 5 inches (127.0 mm).

FSC 6145

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:52Z
Check the source to verify that this is the current version before use.

Requirement [1]	M22931/09 dash number			
	-001	-002	-003	-004
Inner conductor	Copper wire or tube	do.	Copper wire	Do.
Overall diameter (±.002(.05))	0.161(4.09)		0.165(4.19)	
Dielectrics				
Cable core	Polystyrene		Polyethylene	
Insulating layer	Polystyrene		Not applicable	
Outer conductor diameter				
Inside (nominal)	0.421(10.69)		0.456(11.58)	
Outside	0.500(12.70)		0.530(13.46)	
Jacket	Not applicable	Polyethylene	Not applicable	Polyethylene
Overall diameter [+.020(.51), -.010(.25)]		0.575(14.61)		0.615(15.62)
Min. wall thickness		0.035(.89)		0.035(.89)
Insulation defects, spark	Not applicable	Applicable	Not applicable	Applicable
Attenuation (dB/100 ft. max)				
30 MHz	0.48	do.	0.46	Do.
400 MHz	1.93		1.93	
3,000 MHz	5.90		5.90	
Velocity (percent)	86.5		85.5	

[1] Dimensions are in inches; millimeters are in parentheses.
Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:52Z
Check the source to verify that this is the current version before use.

NOTES:

1. Cross reference information is as follows:

Part number	RG number	Former classification Class	Type
M22931/09-001	RG-236/U	N	I
M22931/09-002	RG-237/U		
M22931/09-003	RG-252/U		
M22931/09-004	RG-253/U		

2. Figures 1 and 2 are for information.

Custodians:
Army - EL
Navy - EC
Air Force - 11

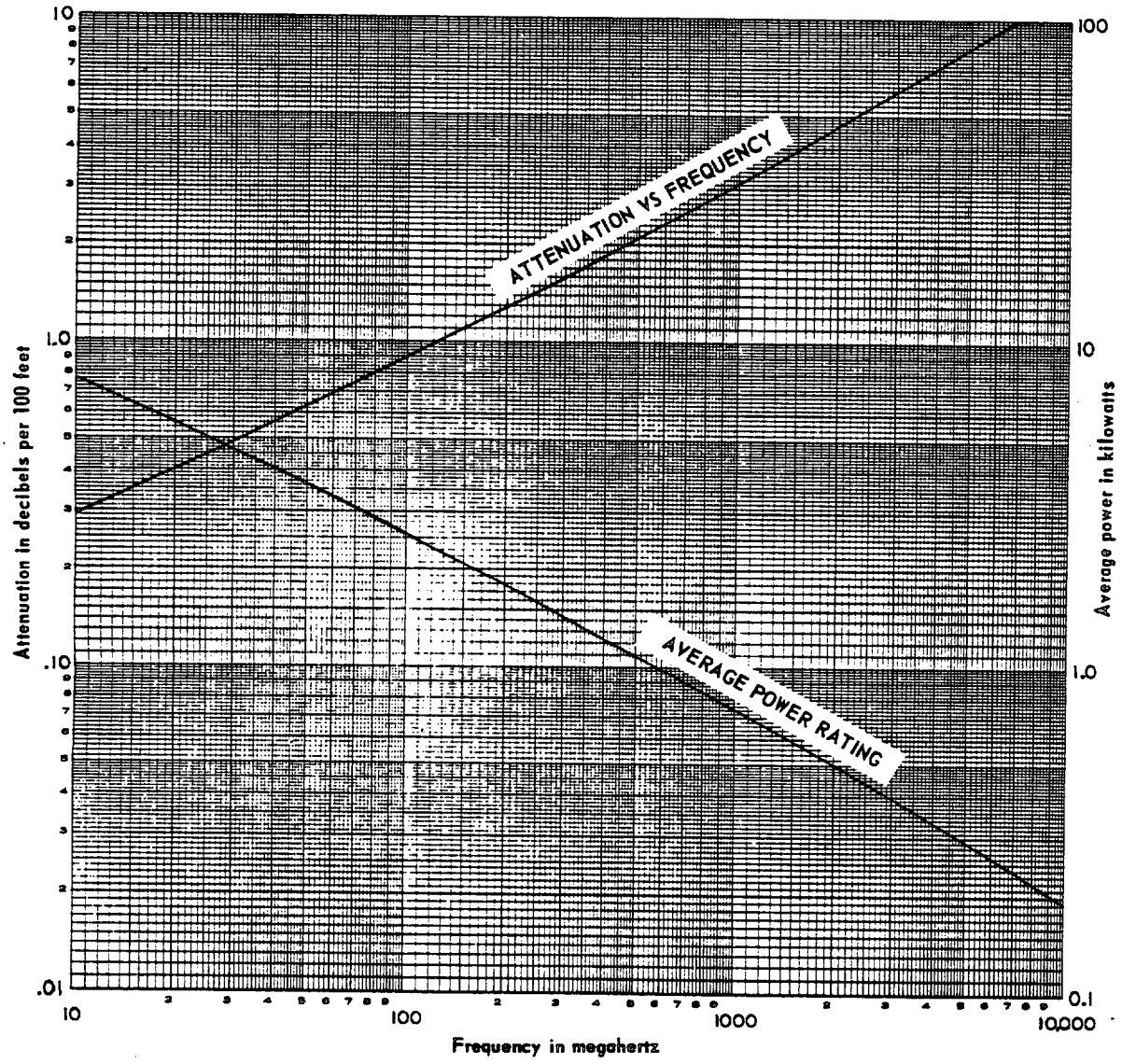
Review activities:
Army - EL, MU
Navy - SH, EC
Air Force - 11, 17, 85
DSA - ES

User activities:
Army - ME, AT, AV
Navy - AS, OS, MC
Air Force - 19

Preparing activity
Navy - EC

(Project 6145-0430-1)

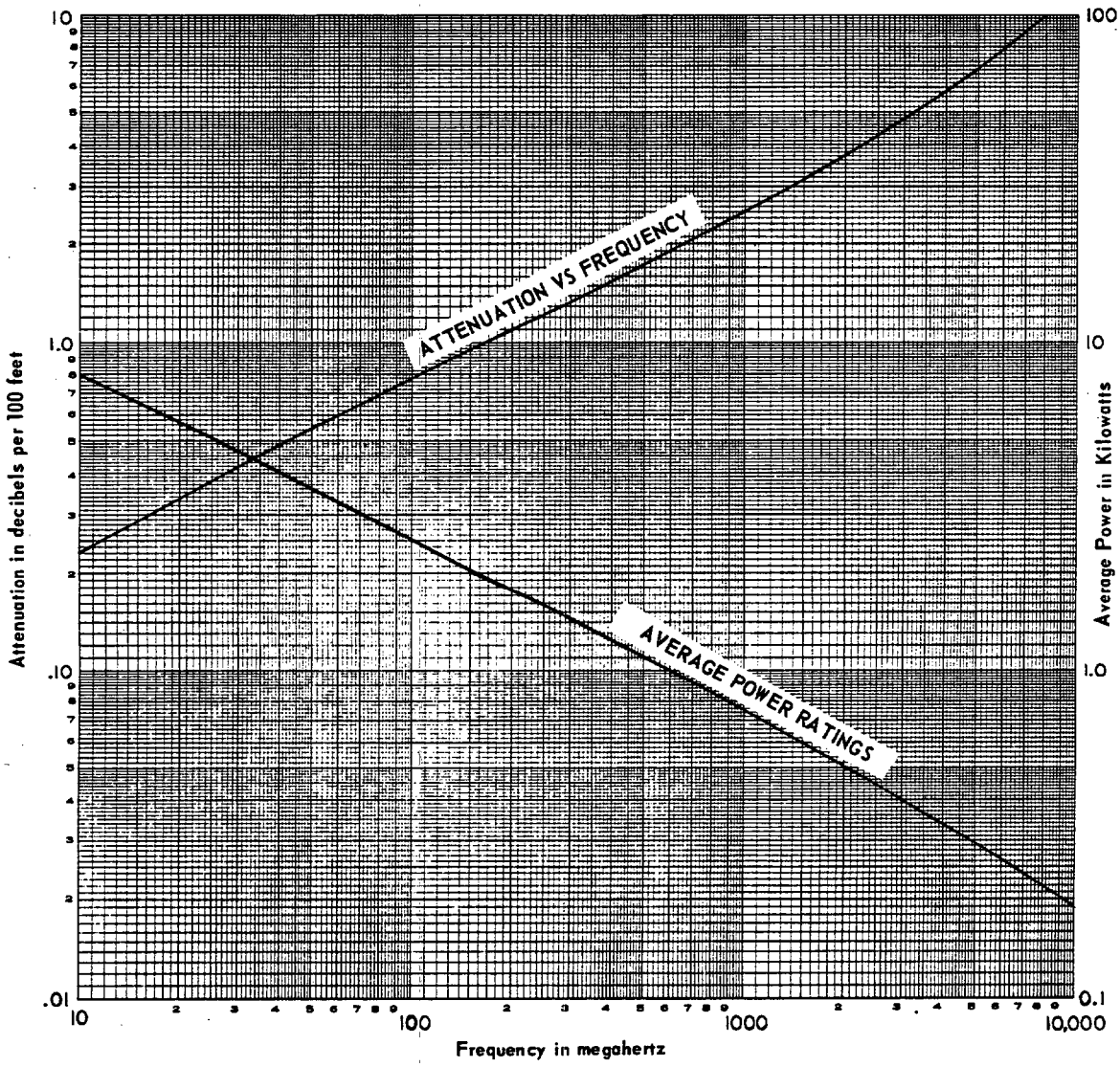
Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:52Z
Check the source to verify that this is the current version before use.



Power rating based upon a 40°C ambient with a 45°C rise in center conductor temperature.

FIGURE 1. Attenuation and power rating curves for –001 and –002.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:52Z
Check the source to verify that this is the current version before use.



Power rating based upon a 40°C ambient with a 45°C rise in center conductor temperature.

FIGURE 2. Attenuation and power rating curves for –003 and –004.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:52Z
Check the source to verify that this is the current version before use.